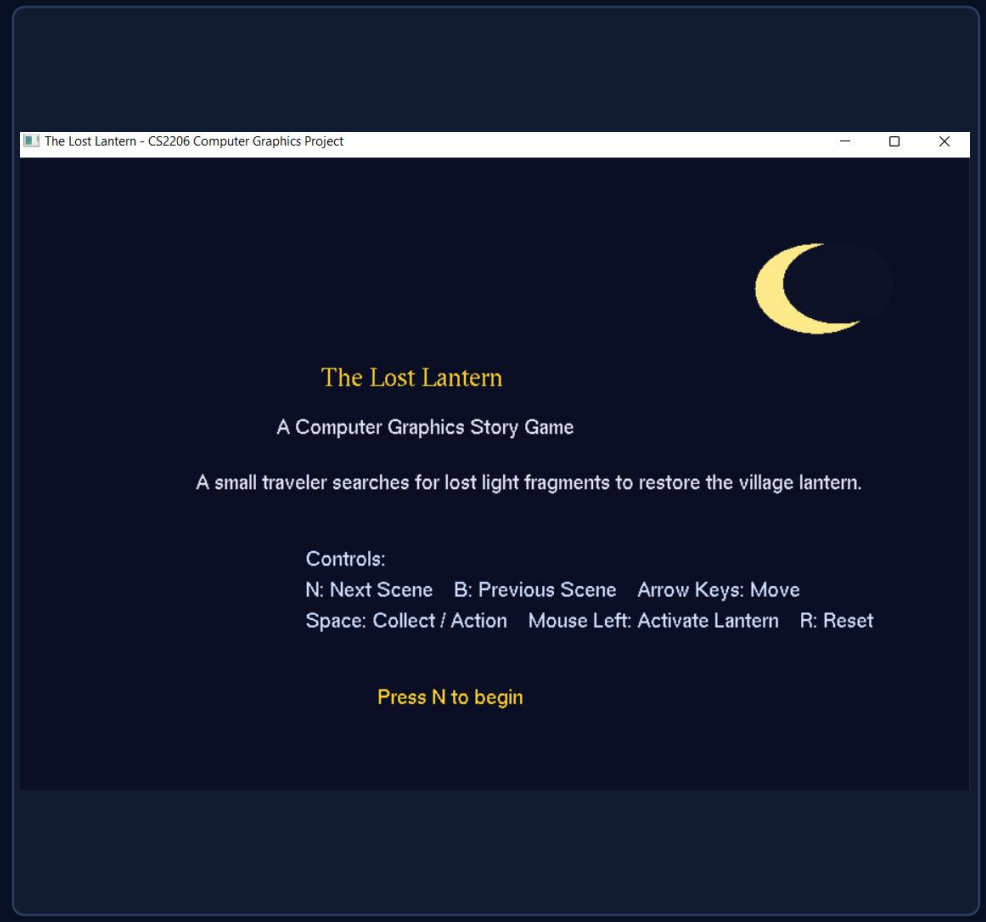
# The Lost Lantern

## A Computer Graphics Story Game

CS2206 Computer Graphics Project

**Project Type: Story Game**



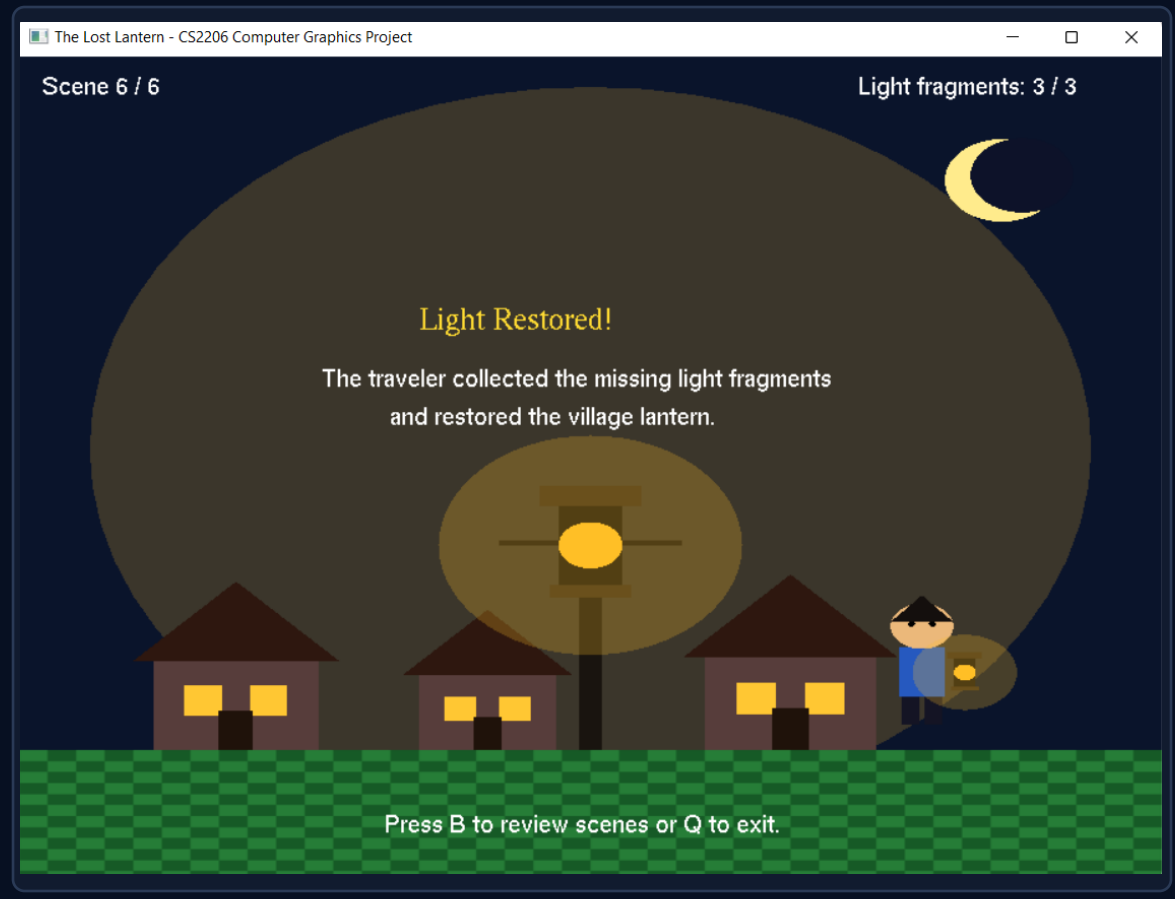

Instructor: Dr. Abi Faisal Mandoura

# Project Idea

A clear story objective with interactive scenes

The Lost Lantern is an interactive story game where a small traveler collects three missing light fragments to restore the village lantern.

- The player moves through six scenes.
- The main objective is to collect 3 light fragments.
- The final scene shows the village light restored.




Final result scene

# Project Requirements Coverage

The implementation aligns with the required CG project rubric

| | |
|---|---|
| **Game / Story** | Story Game with a clear objective |
| **Multiple scenes** | 6 scenes from start to final result |
| **Drawing** | Houses, trees, bridge, tower, lantern, hero |
| **Transformations** | Translation, Rotation, Scaling |
| **Interaction** | Keyboard + mouse actions |
| **Texture** | Grass-like and stone-like textures |
| **Text** | Scene labels, controls, progress counter |

**6**
Scenes

**3**
Light fragments

**2**
Textures

**2**
Input methods

The project is designed to be simple to present, easy to test, and directly connected to core Computer Graphics concepts.

# Story Flow

The player moves from darkness to restored light

1 → 2 → 3 → 4 → 5 → 6

Start Screen | Dark Village | Forest Path | Windy Bridge | Lantern Tower | Final Scene



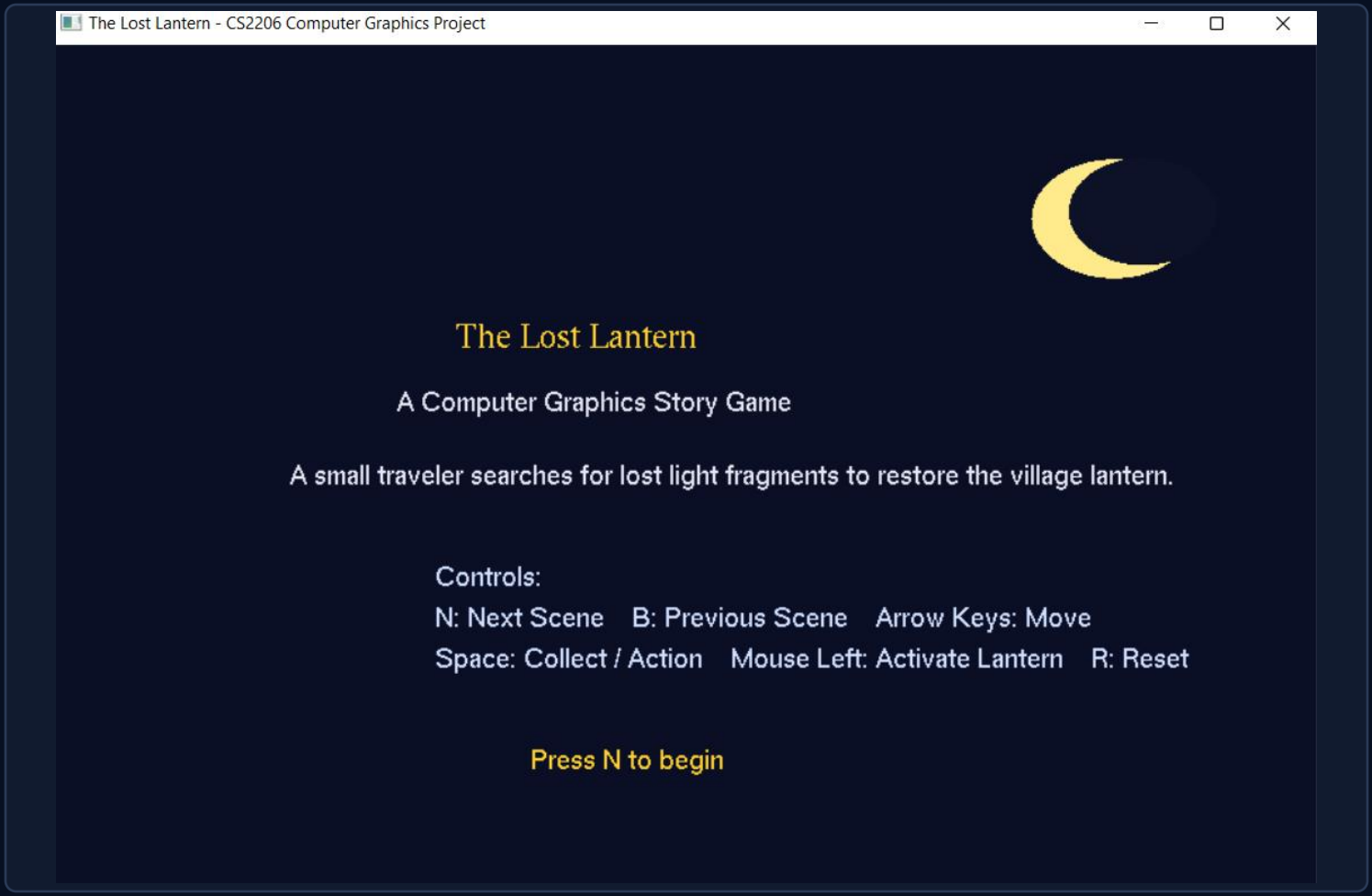






# Scene 1: Start Screen

The player understands the story and controls before starting



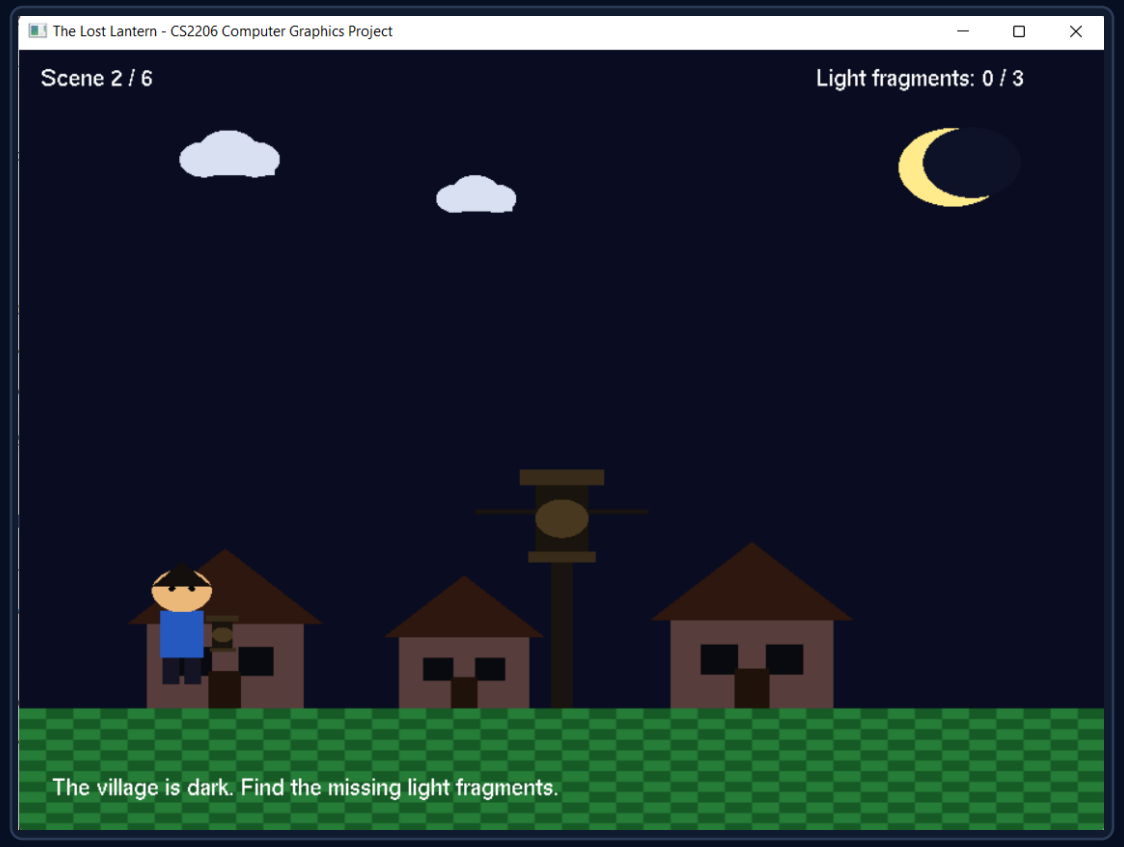

## Purpose

- Introduces the project title and story type.
- Shows the main objective: restore the lantern.
- Lists all user controls before gameplay.

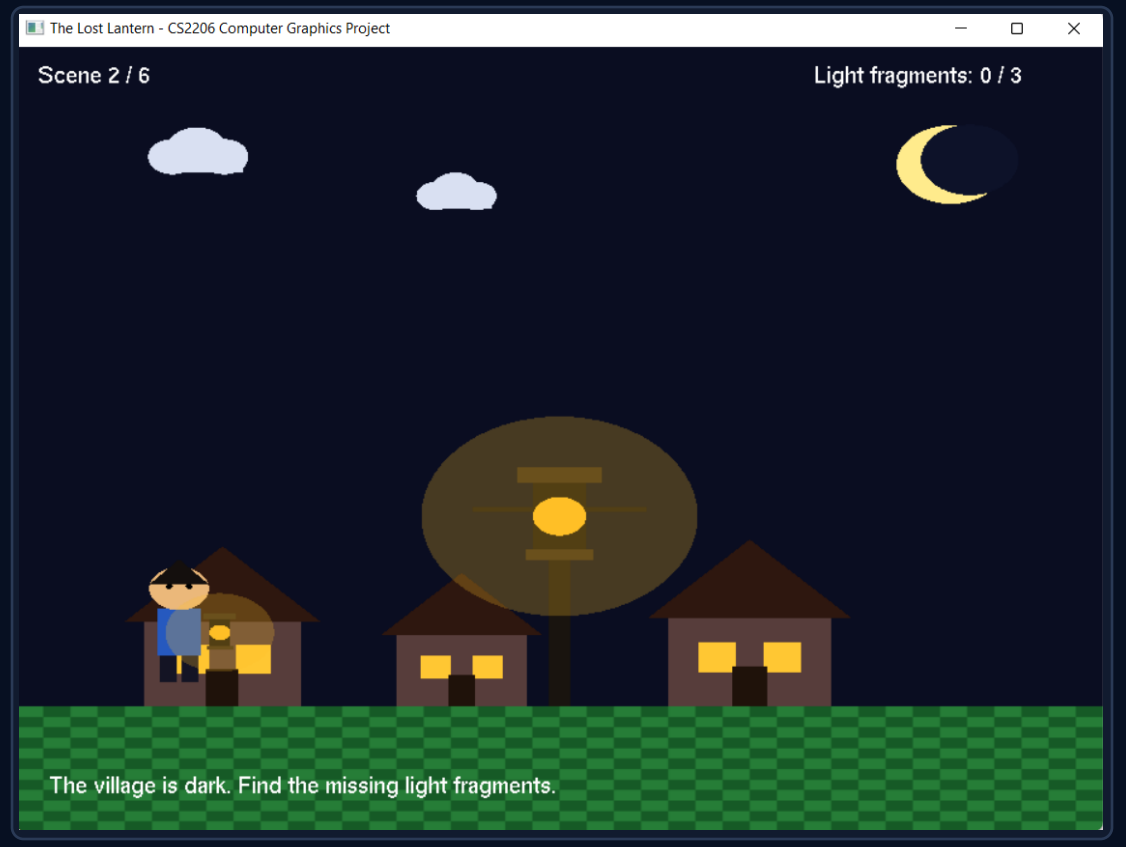
*Presenter line: "The journey begins by pressing N."*

# Scene 2: Dark Village + Mouse Interaction

The problem is introduced, then the player activates the lantern



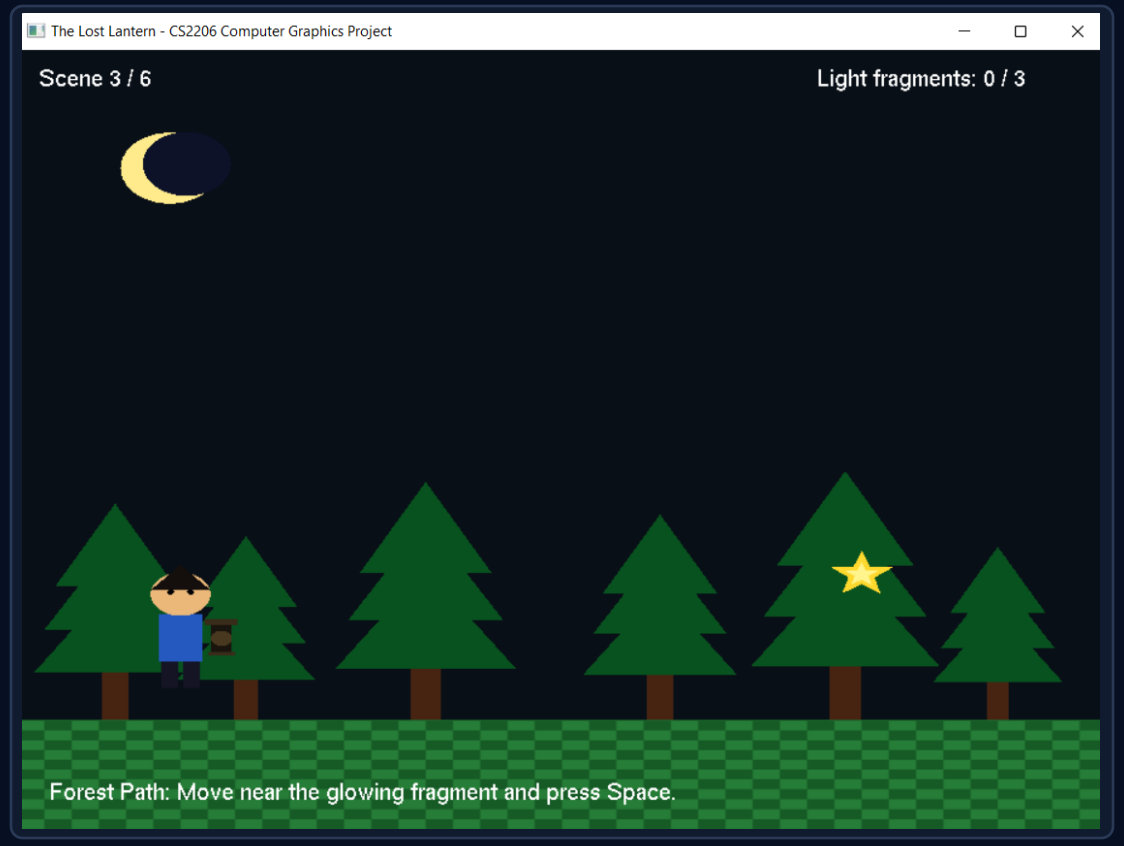

Before mouse click: lantern is off

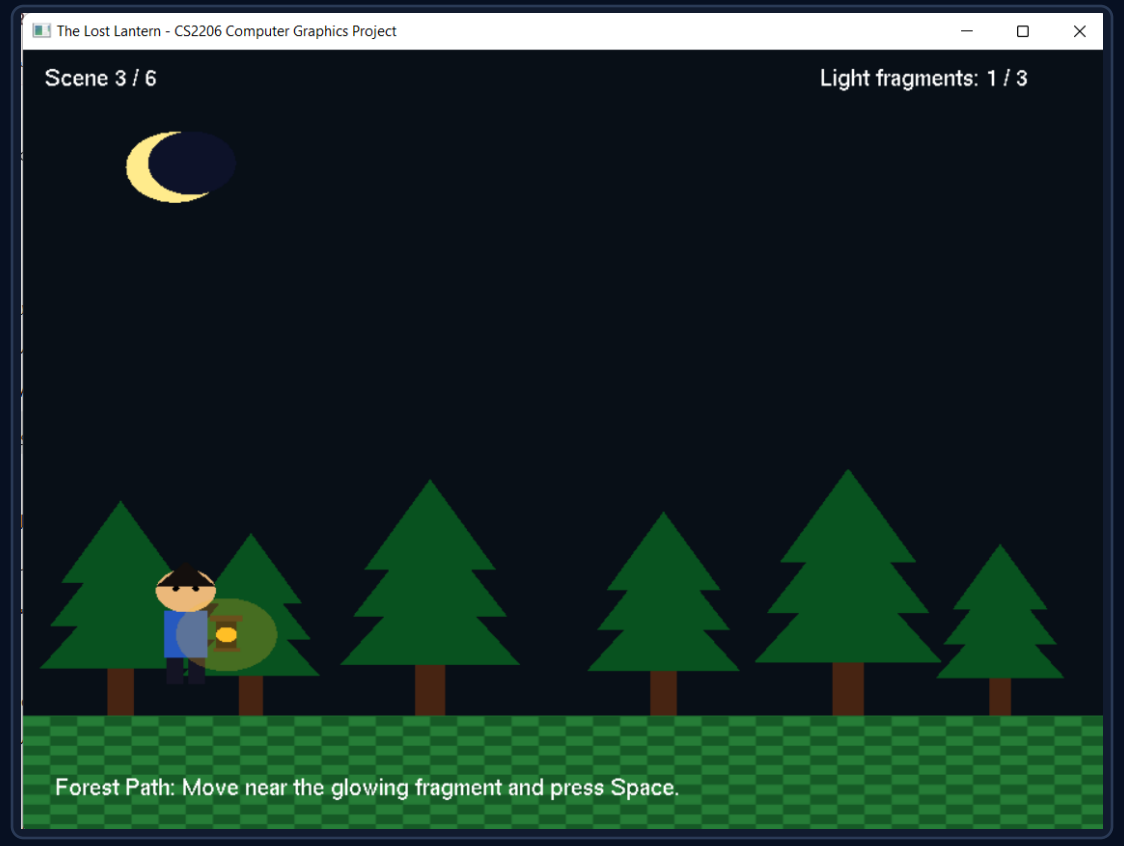



After left mouse click: lantern is activated

# Scene 3: Forest Path

Collecting the first light fragment using the Space key



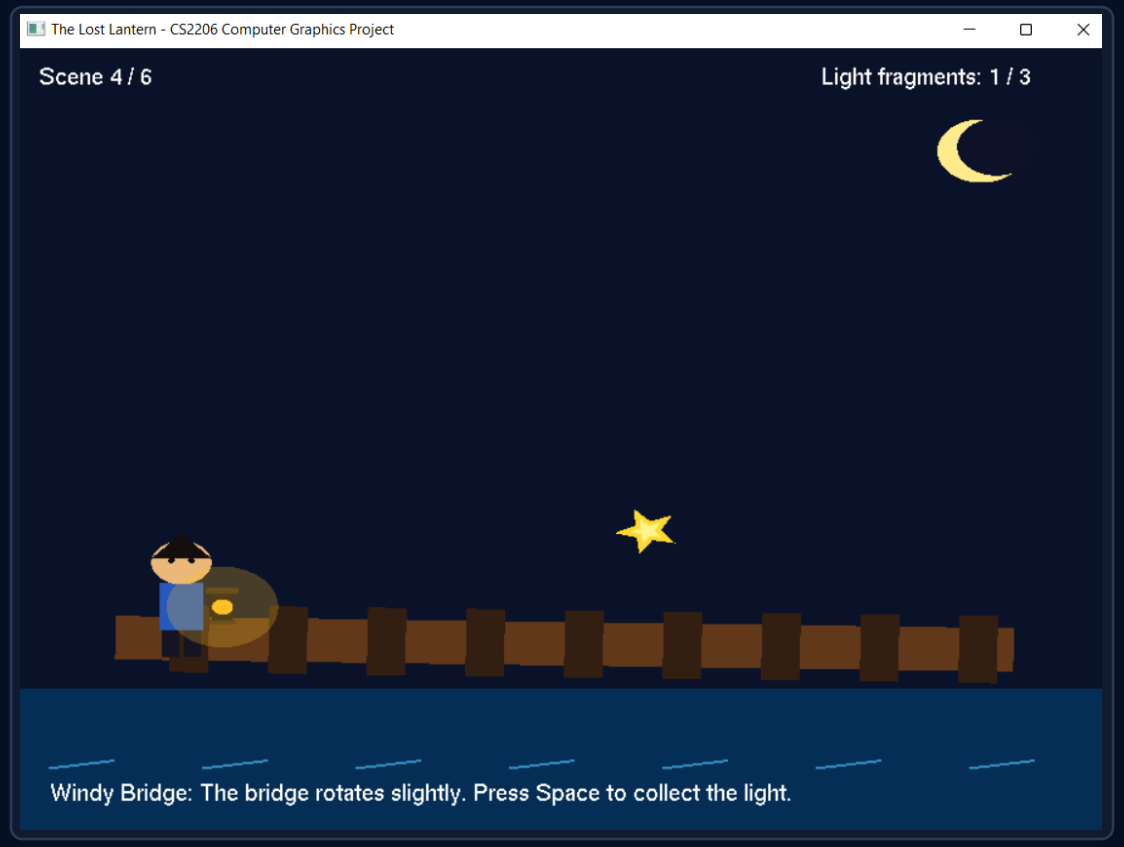

Fragment visible: 0 / 3

Fragment collected: 1 / 3

Concepts: keyboard input, object state update, rotating light fragment.

# Scene 4: Windy Bridge

Collecting the second light fragment while the scene animates







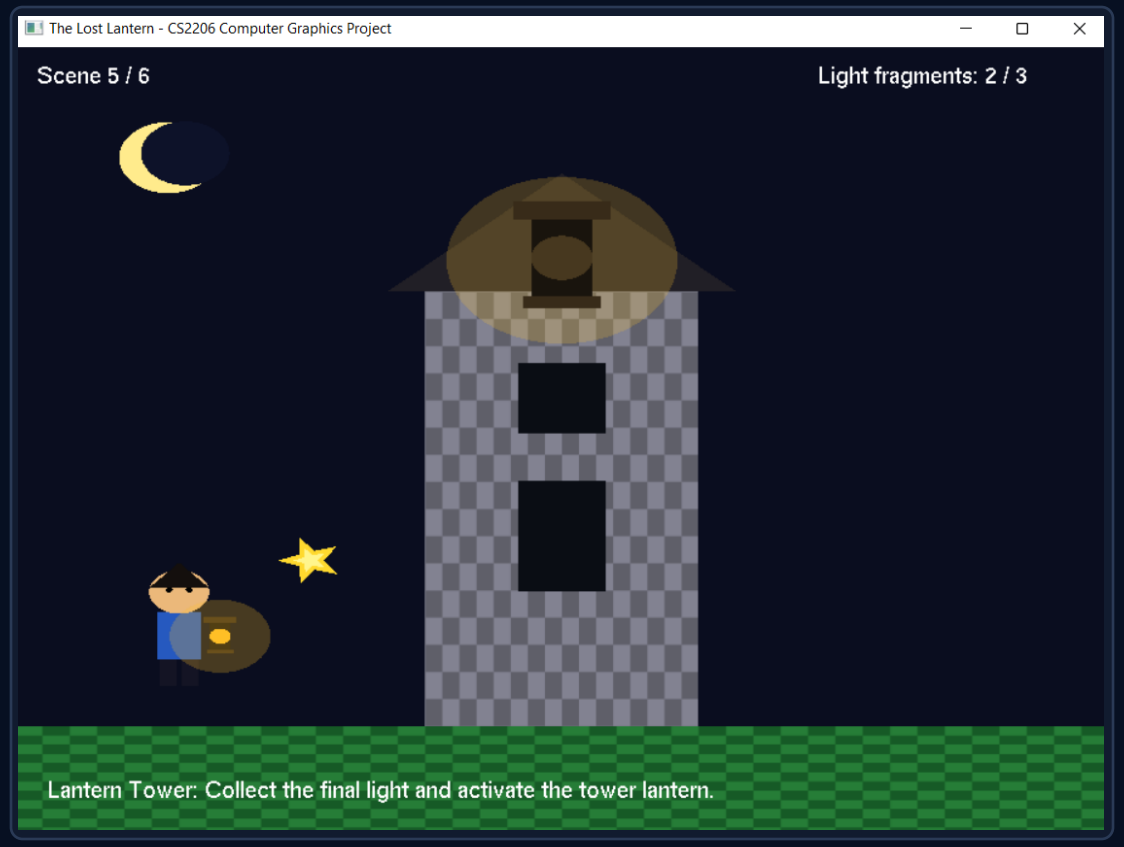
Bridge + water wave animation

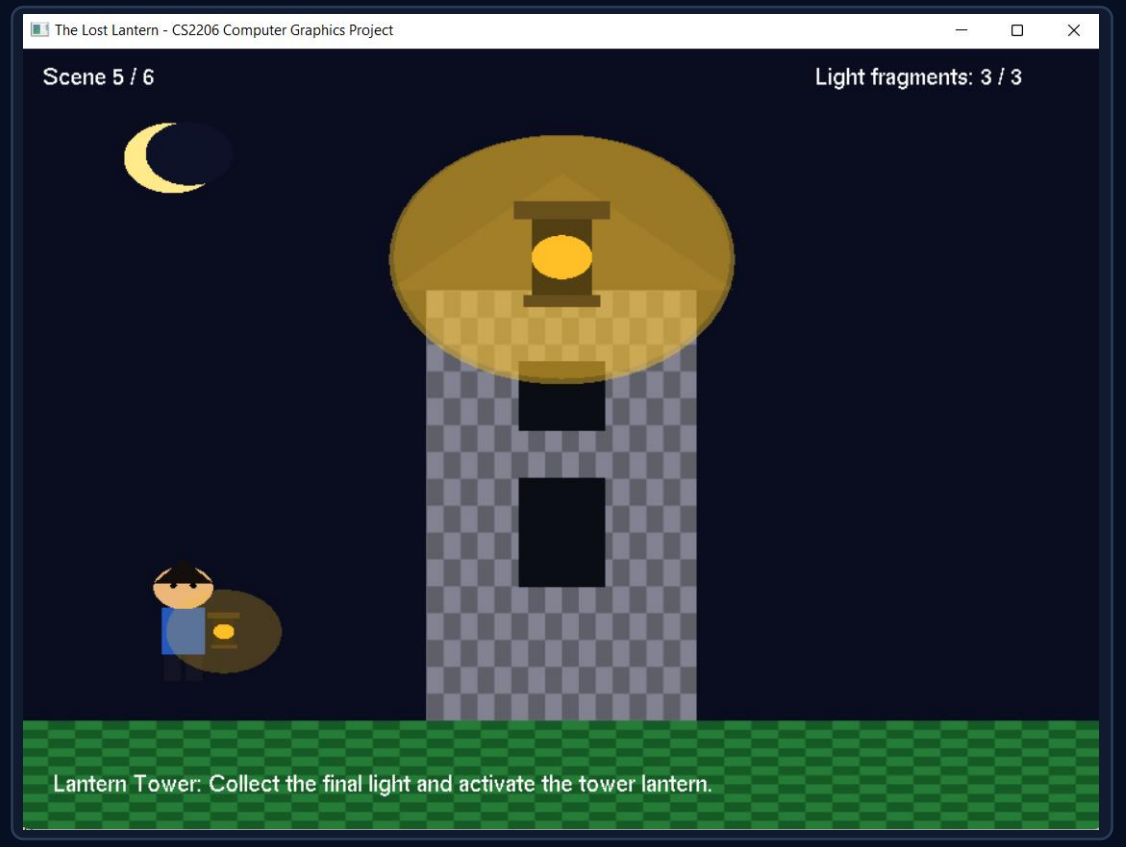
Counter updated: 2 / 3

Concepts: rotation effect, translation-like wave movement, keyboard action.

# Scene 5: Lantern Tower

Final fragment collection and texture mapping



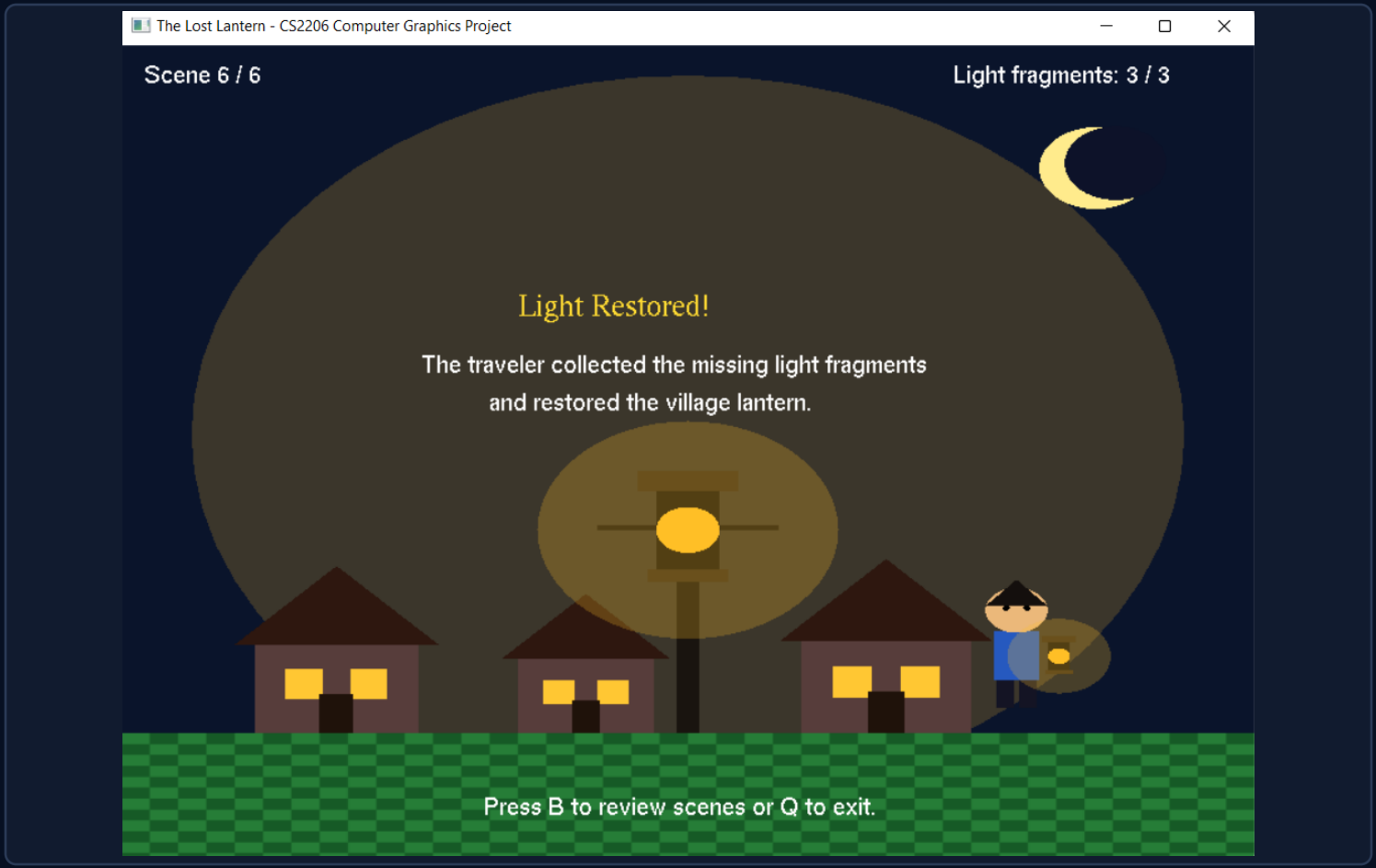

Stone-like texture on tower

Concepts: texture mapping, glowing effect, progress state update.

All fragments collected: 3 / 3

# Scene 6: Final Result

The light is restored after the player completes the objective




## Result

- The player collected all three fragments.
- The village lantern is restored.
- Houses become brighter in the final scene.

*Presenter line: "This scene confirms that the story objective was completed successfully."*

# Computer Graphics Concepts Applied

How the project maps to course concepts

| | |
|---|---|
| **2D Drawing** | Houses, trees, hero, moon, bridge, tower, lantern |
| **Translation** | Hero movement and moving clouds |
| **Rotation** | Light fragments and bridge swing effect |
| **Scaling** | Lantern glow grows and shrinks |
| **Textures** | Grass-like ground and stone-like tower |
| **Interaction** | Keyboard controls and mouse click event |
| **Text Rendering** | Instructions, scene number, progress counter |

# Conclusion and Future Work

The project meets the required functionality and can be extended further

## Conclusion

The Lost Lantern successfully presents an interactive computer graphics story game. It includes multiple scenes, 2D object drawing, geometric transformations, animation, keyboard and mouse interaction, texture mapping, and text rendering.

## Future Work

- Add sound effects and background music.
- Add collision detection and scoring rules.
- Add smoother character animation.
- Use more detailed image textures.
- Add more story levels.

# Thank you

Questions?